Data Extracts, Joins Transcript

Good morning, hello everyone, we will now continue our learning journey for the course Data Visualisation Storytelling, we will be now continuing our understanding of leveraging different features in the Tableau environment, especially how to interact with the data sets and try to build the joints and data blending exercises. This is a very important step in stabilising your data source so that you can perform appropriate calculations and visualisations and build interactive dashboards that clearly communicate the information to your audience. So as I was saying in the last class, preparing your data source or the initial data set is a very important step in any data visualisation exercise.

In this case, you will be using certain features in the Tableau environment to achieve that functionalities. Particularly, we'll be seeing about the relationships, joints and finally the data blending. These tools or techniques will help you to connect the different data sets with common identifiers and also use them simultaneously for building visualisations, thereby bringing more interactivity and cross functionality among different data sets, simultaneously exploiting the trends across data sets to showcase the insights to your audience.

So let's start with the Tableau public environment. So by now you should be familiar with logging into the Tableau public environment. I have logged in using my account.

Then I'm going to do a web authoring, which is the app based version and it is done online. And I have downloaded one particular data set called the bookshop data set, which is available from here. I will post the link in the class materials or else you can also search from this particular bookshop data set Tableau, you will get the direct link to this article where you are expected to download this particular Excel file bookshop.xlx. So once you download that, it will be there in your downloads or in your local drive.

From there, you can simply drag and drop to the upload section and the file is getting uploaded. As a good practise, you can always try to first scan the data on your own and then you can work about the features in the Tableau environment. So if possible, if you are dealing with an Excel file or a text file, it is a good habit or a practise to understand the underlying data by just eyeballing the raw data.

Sometimes if you are working with certain tables or formats which are internal to your processes and system, you might not be able to visualise data. But still, in case if you have a chance, please do try to see the different types of what are the data and what is it containing. So now this is a data about book author and book author sales.

So you have author ID, book ID, and quickly if you are coming to sales, you have the order ID and item ID, series. Then there are different book IDs and ratings of the publisher. There are many different fields, but the main point that we are trying to understand is this is the same data about books, sales, ratings and reviews, but organised in different sheets.

And our task here is to connect them, connect individual sheets so that we can perform combined operations across the sheets. So this is the task for today or in this lesson. And we will try to achieve that by using some special functions in Tableau, especially the join and data blending.

And we will take it from there. Now let us go into the Tableau environment. So you have loaded all the sheets and this is the physical layer or the where your data source is getting built up, OK? So these are the different sheets.

So you can try to select a couple of sheets and then see how they are interacting. So let us first start with the author sheet and just selecting it. Author sheet has come here.

I'm just clicking the update. So you can see the number of fields that are there, author ID, first name, last name, birthday, country of residence, hours of writing per day, HRS is hours of writing per day. So you have these details here, which is this is the only numerical value here.

This is the date value here, OK? And which are again, you have the author's last name, then author ID. So now let us try to see what is that we want to connect with. Go back here.

Look at the author. This is the author info. This is what we have been seeing.

Then you have the book ID. So here you don't have the book ID, you just have the author ID. In book, there is no, there is author ID, sorry, there is author ID and these are the books written by the author.

Now let us try to connect these two sheets, OK? So I'm just taking the author and book. Let us explore other things later. So this is the relationship.

This is what we call as a noodle. And this clearly shows it is connecting automatically. Tableau has detected the primary sheet author sheet is connected to the secondary sheet book sheet where the common identifier is the book.

So this is simply what we call it as a relationship where the Excel sheets are talking to each other using one particular identifier. In this case, the key field in this case, which is the author ID. There might be multiple author fields, multiple key identifiers.

So you can also add them like if you have such, you can also add. So you don't, in this case, you don't have any other field. But if you had any field, which is also there, you can add that.

So it will be sequentially matched and the data set is created. Now let us look at the data set. So this is the title, book ID and author ID, OK? So then once you change the, yeah.

So once you change it to the author sheet, so you click on the book sheet you have here, the book variables, then you have select on author, you have the author sheet. So now the point that I'm trying to make is once you click on this relation, it is showing, OK, how this is related. So you have all the operators, logical operators, how to relate these two and how to connect these two sheets.

So in a relationship, the underlying logic in Tableau is such that it tries to connect these two sheets automatically using the particular ID that is there, which is common in both the sheets. And then you can directly use them in your functions. So now this relationship is established using the author ID.

So which is there. And you can also add this performance options. For example, this is what we call as cardinality principles, whether the mapping is many to many.

For example, there is one author and there is one. If it is a unique combination, you can use one to one, OK, or else you can use one to many. One author can map to many books there or many authors can have too many books.

So the default is many to many and it is advisable unless you are very sure how the mapping is working. Don't disturb with this and the system will automatically take care of that. Now, let us try to see how this will help us in working out the different examples.

So what I'm trying to do, I'm going to sheet one. OK, so it is creating an extract. As I was saying in last class, data extract is nothing but a snapshot of your data created for that one particular instance out of your base data.

It helps in faster query so that it will complete faster query and helps in efficient analysis of the data. So now what am I trying to do? I have just have the author ID. So I'm just pulling it.

So this is the author ID. So I can also make it as. So these are the author and I just going to make the count.

So there are about 40 authors that I have. And in fact, if I do the count author ID instead, I want it to be measured to be distinct count. How many unique authors are there? So of course, the unique authors is also 41.

So there are 41 authors. And then I'm trying to see that with the country of residence. OK, so each country of residence you have like this or else I'll take it here.

So you have the country of residence. This is the number of authors country are based on the country of residence. OK, then what am I adding from this sheet? I'm directly taking the book ID.

OK, so so distinct count of books as per the country. And I want to change it into sideways so that I can compare which country has more authors and more books published. So you have all these values coming up right away, very clean.

So instead of having two graphics which could have been created in two separate sheets, you have connected them with the author ID and it is clearly able to measure across the sheets. This way relationship is built by identifying the key variable which is there in the both the sheets. So now what I can also do, I can also get the hours of writing.

So we have the hours of writing also here. So clearly showing if there is hours of writing, number of books and number of authors, we have all these numbers. So you can also try to format, try to include other variables.

So the factor is the main point is you are able to work across the sheets using a common identifier and get more insights. For example, if the sheets are not connected, you will not be able to address the combined question of which country has the highest number of working hours or the writing hours, but relatively low number of authors and publications. So you can answer those questions or present this data to your audience.

So this is how you can do this features in by combining the data. This is particularly what we call it as a relationship. Now let us talk about joints.

So let us talk about joints. So for every sheet, you have to go back and redefine your relationship. So here, what am I doing? I'm trying to, for sheet 2, I'm saying edit data source.

So I'm going here for this sheet, I'm saying edit data source. So I'm going to edit data source. Here I'm trying to, so now this is how you can build relationship between two sheets.

For example, if you are taking the rating sheet. Okay, so it connects sequentially with the rating sheet using the book ID, author will get mapped to book, book will map out to ratings. So once we go to this particular sheet, you also have the, now you have three sheets here.

So what am I doing? I'm just trying to take the country of residence as rows. Okay, then author ID, we can just show the values. These are the country of residence.

This is the author. Then I have, okay, hours of writing. Then I have the ratings also, some of ratings.

So this is what we have. Maybe here it would be appropriate to get the first name of the author. So I'm just removing this.

So for example, you can simply see that I'm not using anything from the sheet called book one, whereas I'm using directly author and rating sheet, but I'm connected via the book sheet. So like this, by building relationship, you can come, you can connect two, three data sheets and work simultaneously for improving your visualisations. We'll now explore the join feature in the Tableau space.

We will now try to understand how we can connect two or more databases for generating a specific data structure. We are, we have seen relationships, which is the default way in which Tableau connects the different data sources. For example, when you combine two sheets using a common identifier, the Tableau using the relationships creates a larger data set, which preserves the features of the original data sets while allowing you to construct the visualisations by exploiting the common features in both the data sets.

However, you can revert back to the original data set without much, without the changes. Now, however, in cases where you use the joint function, you want to create a particular data structure or create a subset of data with specific features of the common element. This is very useful in allowing hierarchy access to the different employees in the business.

For example, you have the transaction data and sales data, and you don't want all levels of employee seeing this data. So therefore, you can create an employee ID with the designation and interact with these two data sets, the transaction, the sales data, depending upon the hierarchy of the employee. In that case, you would want each employee to look at a specific portion of the data set, which can be achieved using the joint functions.

So there are four types of joint functions that are generally used, which are inner joint, left joint, right joint and fuller, full outer joint. And these joints are simple analogous to our set theory that we learn in our high school mathematics. You have A intersection B, which is nothing but the common elements.

Then you have left union, left joint, which preserves the elements from the primary set plus the common elements. Then you have the right joint, which preserves the common elements plus the secondary set. Then you have the full outer joint, which is nothing but A union B, where it takes all the elements and the common elements from both the data sets and tries to create the integrated data set.

Finally, you also have a union, which is not exactly like a intersection or a union function in the set theory parallels, but you have common elements and you keep appending the rows. A best business use case would be tracking the daily sales data. You would have specified number of columns like the transaction ID, time, purchase value, quantity, so on and so forth for a particular day.

And there are about 1000 transactions on a given day. And the next day you have 5000 transactions. You simply append these data together to get the overall sales data and work with them.

In such cases, the columns are common between the two data sets and you just append the rows to create a larger data set. These are the different ways in joint function is used. Now, let us see a couple of examples to understand how the different types of joints work.

And then we will go to understanding the blending of data. So all this is being taken from the Tableau source itself. So you are advised to refer to this data set again and practise.

So what I am doing here is I have created two more data sheets with partial data from author and book sheets. So this is the same Excel example that we were using earlier, the bookshop data set. So you can go back and work on the same thing.

So what I am doing, I am trying to first understand how book is related to authors. So I have got a book into the canvas or the work area. And I click this particular button to create the joint functions.

Once I have opened this, I will try to drag the author field into this. OK, even before that, I just wanted to show you how many rows are there in the book ID. There are about 58 rows, OK, in the book sheet.

Each representing a different title of the book, author ID, country ID and book ID. So these are the different things that we have here. So I just took and now taking the subset of author here and I just got the author here.

OK, so now I'm just updating it. I just want you to see now the rows have reduced to 37. And what is happening here is performing an inner joint, which means it is taking the common elements, OK, common elements from both the sheets and presenting the data only for them.

OK, it is a presenting the data for common elements which are there in the both sheets. And what is the variable based on the common elements are identified is the author ID. For example, in this sheet, you have the author ID, which is the key which we have used for relationship also.

In the book sheet also, you have the author ID and it is matching the common elements. If you see, these are the common elements AM 329, 329 and these are ordered and altogether you have 37 common elements from both the sheets which are combined and the data set is created. This is one way to join the data.

Now let us explore with the same thing, the left joint, wherein the. Now you have the 58 rows, OK, wherein the data is populated. So you please see the data is now populated for the left sheet along with the common elements.

So first you have the common elements, the author ID, OK, the author ID, common elements have been presented 37 rows. After that, you still have the original data or the primary data, in this case, the book sheet and all other null values are replicated. So this is how your data is getting created.

So you are trying to look at the common elements from the book data and the author data while preserving the all the elements from the book data. This is a left joint. Now we can do the opposite.

We can do the right joint also, wherein you would take only the common elements and the full data from the author sheet. So look at, you should get, you are getting 43 rows where the elements which are not there in the book ID are represented with null values, but all the elements in the author's data set are represented. So these particular authors are not present in the book ID, therefore it is not creating the data for them.

It is representing as null, but it is taking the entire data for the authors and it is giving only partial data or the common elements which are there in the book ID. Then you take the full outer joint, which is the final one, wherein you have the entirety of data from both the sheets. OK, so you have the author ID for which there are no corresponding book IDs.

Similarly, when you go down, you have the book IDs for which there are no author IDs. So this is how your full outer joint works. And you can also, if you have two or more matching variables, you can give more conditions here.

OK. And you can also define joints not only as equivalents, but as inequivalents or being greater than. However, be very, very careful on how do you use joints because they will modify the data once for all.

And once this is done, you just do the create extract. So it will create the extract and just using the full union data. So now this data is created, which will be used for visualising.

OK, so this is how you can do this particular extraction is complete. So extract contains all data. So this is what has been added here.

So now you go here, then you can create. So author ID as rows. OK, then you have the maybe you can just read the nationalities.

You just got this. OK. So you can do this nationality wise number of authors which are there.

And you don't have authors from Singapore and Netherlands, which are in this particular datasets. So that's how you can use the joint function to create subsets and work with the data. Now we will now look at another.

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